

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Zoology**Paper: ZOB-502 — Biochemistry & Mammalian Physiology**

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in both sections is compulsory. Use a separate answer book for each Section. The figures in the right-hand margin indicate marks.

SECTION – A**(Biochemistry) (Marks: 35)**

Note: Answer three questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

(a) Indicate whether the following statements are True or False. Justify your answer precisely:
[4 × 2 = 8]

- (i) Formation of ES complex is a compulsory step of enzyme catalysis.
- (ii) Cholesterol is composed of more than one fatty acids.
- (iii) One DNA Pol-III complex catalyzes DNA synthesis simultaneously at both strands of the parental DNA.
- (iv) 'Shine Dalgarno' sequences determine initiation of prokaryotic transcription.

(b) Define the following: [3 × 1 = 3]

- (i) Genetic Vector
- (ii) pI of an amino acid
- (iii) Acid-base catalysis.

Question 2. Answer the following parts: [6 + 6 = 12]

- (a) Illustrate stepwise formation of transcription initiation complex for RNA pol II activity.
- (b) Describe (in diagrammatic form only) four step events of chain elongation during prokaryotic protein synthesis.

Question 3. Answer the following parts: [4 + 6 + 2 = 12]

- (a) Write the reactions producing FADH₂ and GTP during the TCA cycle.
- (b) Illustrate arrangement of ETS at inner mitochondrial membrane.
- (c) Construct LB-plot of the enzyme kinetics.

Question 4. Write short notes on the following:

[3 × 4 = 12]

- (a) Prostaglandins
- (b) tRNA
- (c) Restriction endonucleases.

SECTION – B
(Mammalian Physiology) (Marks: 35)

Note: Answer three questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

- (a) Select and write the most suitable answer for the following: [2 × $\frac{1}{2}$ = 1]
- (i) Muscle contraction consists of the cyclic attachment and detachment of the:
- (1) globular head portion of myosin to the F-actin filament
 - (2) tropomyosin to G-actin
 - (3) myosin head to G-actin
 - (4) ATP to actin-myosin complex
- (ii) 2,3-diphosphoglycerate is a charged anion which binds to:
- (1) α -chains of deoxyhaemoglobin
 - (2) β -chains of deoxyhaemoglobin
 - (3) α -chains of oxy-haemoglobin
 - (4) β -chains of oxy-haemoglobin
- (b) State whether the following statements are True or False. Write brief justifications for your answers: [2 × 3 = 6]
- (i) Walls of afferent arteries contain rennin-secreting juxtaglomerular cells.
- (ii) Between trachea and alveolar sacs, airways divide 20 times and the first 16 generations of passages form the conducting zone of the airways.
- (iii) In human, Hering-Breuer reflex is not activated until tidal volume increases to greater than around 1.5 L.
- (c) What is the speed of conduction if the distance between cathodal stimulating electrode and exterior electrode is 4 cm and the latent period is 2 ms? [2]
- (d) Define the following: [2]
- (i) Haemostasis
 - (ii) GFR.

Question 2. Answer the following parts: [6 + 6 = 12]

- (a) Describe the role of Ca^{2+} during muscle contraction with suitable diagrams.
- (b) Describe the role of pancreatic juice in digestion of proteins and carbohydrates.

Question 3. Answer the following parts: [6 + 6 = 12]

- (a) Describe the role of respiratory muscles in breathing.

- (b) What are the different types of synapse? Describe the process of nerve impulse propagation in a myelinated nerve-fiber.

Question 4. Write short notes on the following:

[3 × 4 = 12]

- (a) Action potential
- (b) Sarcomere
- (c) Chloride shift.

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